

■ Roadmap for LLMs & NLP (Python Only)

1. Python Skills (Quick)

Basics: loops, functions, OOP, file handling

Libraries: NumPy, Pandas, Matplotlib/Seaborn

■ Since you already know JS, this will be fast

2. NLP Basics

Text preprocessing: tokenization, stopwords, stemming, lemmatization

Libraries: NLTK, spaCy

Example: clean and tokenize a text dataset

3. Embeddings (Core for LLMs & RAG)

Concept of word embeddings (Word2Vec, GloVe) → just theory

Modern: Transformers embeddings

Libraries: sentence-transformers, HuggingFace Transformers

■ Know how to get embeddings and store them in a vector database

4. Transformers & LLMs

HuggingFace Transformers basics: load models, tokenizers, inference (text generation)

Fine-tuning: LoRA (Low-Rank Adaptation)

HuggingFace Trainer API

5. RAG (Retrieval-Augmented Generation)

Workflow: User Query → Embeddings → Vector DB → Relevant Context → LLM Answer

Tools: LangChain, LlamaIndex

Vector DB: Pinecone, Weaviate, Qdrant, FAISS

Example Project: Upload PDFs, Store embeddings in DB, Query with an LLM

6. Serving & Deployment

FastAPI/Flask → wrap your model as API

Dockerize chatbot

Deploy to cloud (AWS/GCP/Render)

7. Projects to Build (Step-by-Step)

Simple text summarizer with HuggingFace

Q&A; bot using embeddings + FAISS

RAG pipeline with LangChain + OpenAI/LLaMA

Deploy chatbot on web (React frontend + FastAPI backend)